

COMPETENT PERSON FALL PROTECTION – STUDENT GUIDE

A Competent Person is required on job sites to recognize fall hazards and choose appropriate fall protection systems for workers.

A **Competent Person** under WAC 296-880-095 is defined as “An individual knowledgeable of fall protection equipment, including the manufacturer's recommendations and instructions for the proper use, inspection, and maintenance; and who is capable of identifying existing and potential fall hazards; and who has the authority to take prompt corrective action to eliminate those hazards; and who is knowledgeable of the requirements contained in this chapter regarding the installation, use, inspection, and maintenance of fall protection equipment and systems.”

Some fall protection systems use the term **Qualified Person**.

Qualified person is defined as an individual “who, by possession of a recognized *degree, certificate or professional standing, or who by extensive knowledge*, training and experience, has successfully demonstrated his ability to solve or resolve problems relating to the subject matter, the work, or the project.”

TRAINING REQUIREMENTS

Competent and qualified person training should include all forms of fall protection systems including the manufacturer's recommendations and instructions for the proper use, inspection, and maintenance. WAC 296-880-10005

- The nature of fall hazards in the work area;
- When fall protection is required;
- What fall protection is required;
- The correct procedures for erecting, maintaining, assembling, disassembling, and inspecting the fall protection systems to be used;
- The use and operation of fall protection systems;
- Limitations of fall protection systems used;
- Proper care, maintenance, useful life, removal from service; and
- The requirements of this chapter.

Make sure before an employee is allowed to perform work requiring the use of fall protection that the employee can:

- Demonstrate an understanding of the training specified above; and
- Demonstrate the ability to use fall protection properly.

RETRAINING

When the company has reason to believe that any affected employee who has already been trained does not have the understanding and skill required by of this section, you must retrain each such employee. Circumstances where retraining is required include, but are not limited to, situations where:

- Changes in the workplace render previous training obsolete
- Changes in the types of fall protection systems or equipment to be used render previous training obsolete
- Inadequacies in an affected employee's knowledge or use of fall protection systems or equipment indicate that the employee has not retained the requisite understanding, skill, or motivation to use fall protection.

WALKING WORKING SURFACES

All surfaces on which employees will be working or walking on are structurally sound and will support them safely prior to allowing employees to work or walk on them.

REGARDLESS OF THE HEIGHT

Employers must protect workers from falls when they workers are exposed regardless of the height:

- Floor holes (2 inches in any direction or more)
- Skylights
- Over or near dangerous equipment

Floor holes, floor openings, skylights, or dangerous equipment shall be guarded by either a:

- Guardrail with toeboards on all exposed sides
- All other covers shall be capable of supporting, without failure, at least twice the weight of employees, equipment, and materials that may be imposed on the cover at any one time.
- All covers must be secured against displacement and be marked with the word “cover” or “hole”.

When the guardrail, cover, or hatch is removed another form of fall arrest or fall restraint that meets the standard can be used.

RAMPS RUNWAYS AND WALKWAYS

- Ramps, runways, and inclined walkways that are four feet or more above the ground or lower level must be equipped with a standard guardrail system or the equivalent
- Runways used exclusively for special purposes may have the guardrail on one side omitted where operating conditions necessitate such omission, provided the falling hazard is minimized by using a runway not less than eighteen inches wide.

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WORKING OVER WATER

Personal flotation devices do not eliminate the need for Fall Protection when there is a 4' or greater fall hazard to the surface of the water.

A harness worn in fall restraint eliminates the need for PFDs if the worker cannot fall into the water.

FALL PROTECTION HAZARDS

All surfaces workers stand or walk on must have the strength and structural integrity to support employees safely.

The following types of surfaces and conditions with a 4' fall hazard require fall protection measures by the employer:

- Walking-working surfaces
- Any unprotected sides and edges
- All wall openings that are lower than 39" from floor
- Hazardous Slopes
- Form and rebar work
- Pitched Roofs (greater than 4/12)
- Ramps, runways, walkways including walkways that cross excavations
- Ladderways, floor openings or platforms
- Hatchways and chutes
- Manholes
- Wells, pits, or shafts
- Hoist areas
- Rolling stock for Non-Construction Work

WALKING WORKING SURFACES

All walking working surfaces must be:

- Clean, orderly, sanitary
- Dry if feasible
- Drainage and dry standing places for wet surfaces
- Free of sharp hazards, loose boards, leaks, spills, snow and ice
- Supports the maximum intended load
- Safe access and egress

FALL PROTECTION REQUIREMENT EXEMPTIONS

The fall protection requirements do not apply when employees are on the roof for inspection, investigation, or assessment of workplace conditions prior to the actual start of construction work or after all construction work has been completed. For this exemption:

- No tools
- No stocking of material
- No working is being done on the roof

DOSH also allows an exemption during initial installation of the fall protection anchor (prior to engaging in any work activity), or the disassembly of the fall protection anchor after the work has been completed.

EXEMPTIONS TO THE 4' RULES

- Leading edge. The advancing edge of a floor, roof, or formwork which changes location as additional floor, roof, or formwork sections are placed, formed, or constructed requires Fall Protection at 6'.
- Roofing work. The hoisting, storage, application, and removal of roofing materials and equipment, including related insulation, sheet metal, and vapor barrier work, but not including the construction of the roof deck requires Fall Protection at 6'.
- Erection or placement of structural members.
 - Exception: When the erection or placement of structural members is performed on or from a floor, deck, roof, or similar surface you must use fall protection at 4'. (Note: This does not really seem to be an exemption.)
- Scaffolding
 - Suspended scaffolds and Boatswain's chairs need an additional personal fall arrest system be connected to each worker.
- Excavation in the **affected area**, employees that **are not directly involved** with the excavation process and the employees are **on the protective system** or any other structure in the excavation need fall protection.

HIERARCHY OF CONTROLS

The hierarchy of controls for fall protection, from most effective to least effective includes:

- Elimination, such as using a drone instead of climbing a structure.
- Passive fall protection such as guardrails.
- Fall Restraint
- Fall Arrest
- Administrative systems such as Safety Monitors.

Note: Safety Monitors are listed as least effective, but is interesting to note that an extremely low number of fatalities are linked to administrative systems that were in compliance.

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FALL RESTRAINT VERSUS FALL ARREST SYSTEMS FOR WORKER PROTECTION

Fall restraint is the concept you do not allow workers to go over the edge or fall hazard.

Fall arrest is the concept that the worker may fall but will be caught by a safety system.

Competent persons can choose the system that best addresses the fall hazards on a particular work site.

YOU DON'T WANT TO FALL

Without deceleration devices, a fall in a 'restraint' system can cause thousands of pounds of force on worker bodies. Deceleration devices reduce forces to less than 1800 lbs, but can still result in major injuries.

TRY TO RIG ALL FALL PROTECTION SYSTEMS SO YOU DON'T FALL OFF THE WALKING WORKING SURFACE!

FALL RESTRAINT SYSTEMS

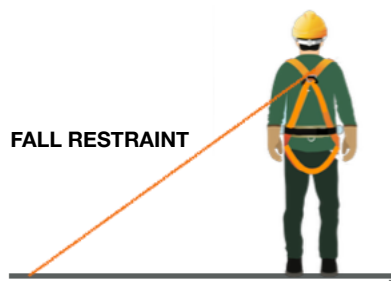
The following are the basic fall restraint systems:

- Personal Fall Restraint
- Guardrail
- Positioning System
- Cover
- Warning Line
- Safety Monitor

PERSONAL FALL RESTRAINT SYSTEMS

Personal fall restraint systems must include:

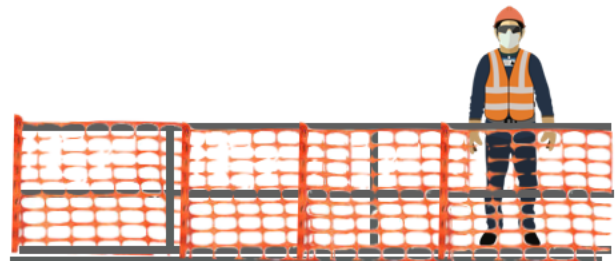
- A full body ANSI Class III harness and self-retracting lanyard.
- Harness must be attached to rigged lines that only allow movement to the edge of the walking/working surface.
- Anchor for fall restraint system must support a minimum tensile load of 4x the intended load.



GUARDRAILS

Temporary guardrail specifications include:

- 39-45" in height with smooth top rail that can sustain 200 lbs force in downward and outward directions.
- Must have a midrail.
- If wood, at least 2"x4" posts no more than 8' o.c.
- Toe board must be at least 3 1/2", no more than 1/4" off surface.
- Additional paneling and protection required if materials might fall off surface.
- 1 1/2" pipe can be used.
- Any material that meets these standards can be used as a temporary guardrail.



Note: When guardrails must be temporarily removed to perform a specific task, the area must be constantly attended by an employee until the guardrail is replaced. The only duty the employee must perform is to warn persons entering the area of the fall hazard.

OVERHEAD PROTECTION

Protect workers below from "Struck by" injuries by:

- Requiring Hardhats
- Signage
- Canopies
- Screens and Mesh able to withhold 200 lbs force from top rail to floor if toe board does not provide enough protection.

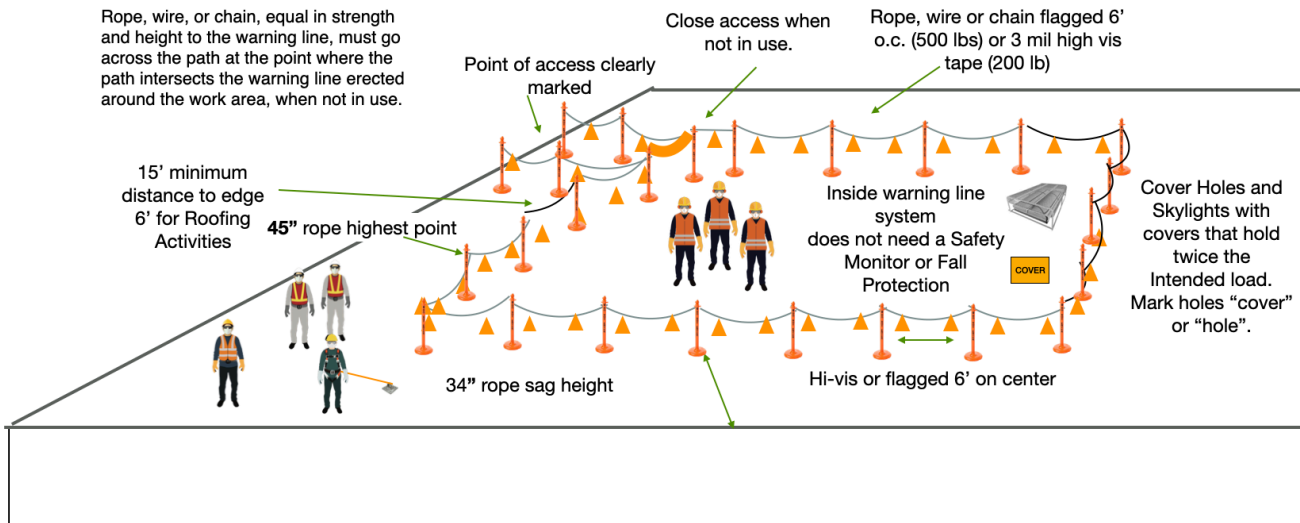
WARNING LINE SYSTEMS

Low Sloped Roofs with pitches 4 in 12 or less

- Roofing work or leading edge work - 6' setback
- Roofing work with Mechanical Equipment - 6' parallel, 10' perpendicular
- Other activities - 15' setback
- Warning lines must be erected around all unprotected sides and edges of the work area.
- Roofing or Leading Edge work outside the WARNING LINE SYSTEM requires a safety monitor or personal fall protection system. Other industries must use Fall Arrest or Restraint.

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WARNING LINE SYSTEM



Note: There are lots of specific requirements for warning line systems. Look up the requirements in WAC 296-880-10015 before setting up a WARNING LINE SYSTEM.

SAFETY MONITOR FOR ROOFING OR LEADING EDGE ACTIVITIES ONLY

DOSH standards slightly supersede OSHA standards for Safety Monitors:

- Be a competent person as defined in WAC 296-880-095.
- Have control authority over the work as it relates to fall protection.
- Be instantly distinguishable over members of the work crew.
- Perform no other duties while acting as safety monitor.
- Be positioned in relation to the workers under their protection, so as to have a clear, unobstructed view and be able to maintain normal voice communication.
- Not supervise more than 8 exposed workers at one time.
- Warn the employee when it appears that the employee is unaware of a fall hazard or is acting in an unsafe manner.

Safety Monitors can be a stand alone fall protection system.

- The warning line is not required when performing roofing work on low pitched roofs **less than 50 feet wide**.
- For information on determining roof widths, see WAC 296-880-500, Appendix A, determining roof widths.

SAFETY WATCH

When one employee is conducting work on a low pitch roof, not within six feet of the edge, and where exposure to falls is infrequent and temporary, employers are allowed to use a SAFETY WATCH system.

- The SAFETY WATCH must be a competent person as defined in WAC 296-880-095;
- There can only be two people on the roof while the safety the one employee acting as the SAFETY WATCH and the one employee engaged in the work activity;
- The employee performing the work must comply promptly with fall hazard warnings from the SAFETY WATCH;
- Mechanical equipment is not used; and
- The SAFETY WATCH is not used when weather conditions cause additional hazards.
- Safety watch must have full control of the work as it relates to fall protection.
- They must have full unobstructed view of worker.
- Must be able to maintain normal voice communication.
- Must perform no other duties while acting as safety watch.

SAFETY WATCH SYSTEM



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POSITIONING SYSTEMS

- Full body harnesses **ONLY!**
- Must prevent a free fall greater than two feet.
- Anchor must be capable of supporting at least twice the potential impact load of an employee's fall or 3,000 pounds, whichever is greater.
- Minimum breaking strength of 5,000 pounds.
- D-rings and snap hooks must be proof-tested to a minimum tensile load of 3,600 pounds.
- Snap hooks must be a locking type.
- Harness with positioning lanyard on hip D-rings or sternal or dorsal D-ring.
- Limits to a 2' free fall, not to be used as Fall Arrest while traveling.
- Minimum 3000 pound anchor.

POSITIONING SYSTEM



FALL ARREST SYSTEMS

- Fall Arrest System
- Catch Platform
- Safety Net

Fall arrest systems must be designed so your “fall distance” does not let you hit the ground. A fall arrest system with a 6' lanyard on the edge of a fall hazard could allow you to fall as far as 18'.

PERSONAL FALL ARREST SYSTEMS

Personal fall arrest systems can have the following configurations.

- ANSI Class III Harness
- Energy Absorbing (deceleration) Device
 - Lifeline with rope grab system that allows 6' free fall
 - Lanyards that allow 6' free fall.
 - SRL that limit free fall to 2' or less.
 - SRL that exceeds a 2' free fall that reduces forces to 1800 lbs or less.
- 5000 lb Anchor

SELF RETRACTING LANYARD



Note: SRLs that limit free fall to 2' or less may use a 3000 lb anchor system.

PERSONAL FALL ARREST SYSTEM DETAILS

- Personal fall arrest systems must have the attachment point on D-ring on center of the wearer's back (dorsal) near shoulder level, or above the wearer's head. (Dorsal)
- It must be a commercial ANSI Class III harness.
- Basic harness design is based on 310 pound weight limit.
- Connecting assemblies must have a minimum breaking strength of 5,000 pounds (22.2 kN).
- Limit maximum arresting force on an employee to 1800 pounds.
- Be adjusted such that an employee can neither free fall more than 6 feet (1.8 m), nor contact any lower level.
- Must limit the deceleration distance to 42".
- Anchor must hold 5000 pounds.
- If using a vertical lifeline, only one worker per lifeline is allowed.

PERSONAL FALL ARREST



CONNECTORS

Connectors for fall restraint and arrest systems, including snaphooks must:

- Be drop forged, pressed or formed steel, or made of equivalent materials, corrosion-resistant finish, and all surfaces and edges shall be smooth.
- Have a minimum tensile strength of 5,000 pounds.
- Opening connectors must be proof-tested to a minimum tensile load of 3,600 pounds, (this must be stamped on snaphook).
- Double locking mechanism.



Never connect carabiners or snap hooks:

- to a D-ring to which another snap hook or other connector is attached;
- directly to a horizontal lifeline; or
- to any object which is incompatibly shaped or that could cause forced roll out
- to another snaphook or carabiner.

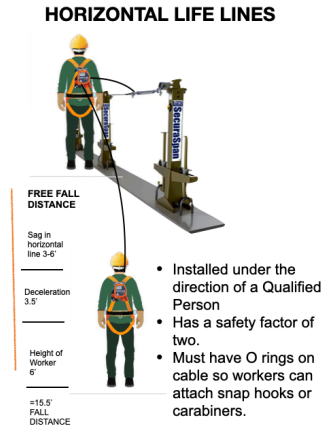


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HORIZONTAL LIFE LINES

The following applies to horizontal life lines:

- Allows multiple workers to use at the same time.
- Reduces some swing falls.
- Works with existing personal fall arrest equipment.
- Needs an O-ring to attach snap hooks too.
- Horizontal lifelines shall be designed, installed, and used, under the supervision of a **qualified person**, and **must maintain a safety factor of at least 2**.



ANCHORS

Anchorage for fall arrest systems must be capable of supporting:

- 3000 lbs. when used in conjunction with a self-retracting lifeline that limits the maximum free fall to 2' or less.
- 5000 lbs for all other personal fall arrest system applications.
- Inspect all anchors for correct number of fasteners, bolts or screws.
- Where anchors are installed, check the concrete or other substrate condition before use.



FALL DISTANCE

Fall Distance can be described as:

- any slack over the fall edge (lanyard, excess rope)
- deceleration device extension (3.5')
- height of the worker (6').

You can fall 9.5' with almost no slack in the line.

Adding a 6' lanyard for allowed 'free fall' means you will drop 15.5', with a safety factor, that is over 18'!

SWING FALLS

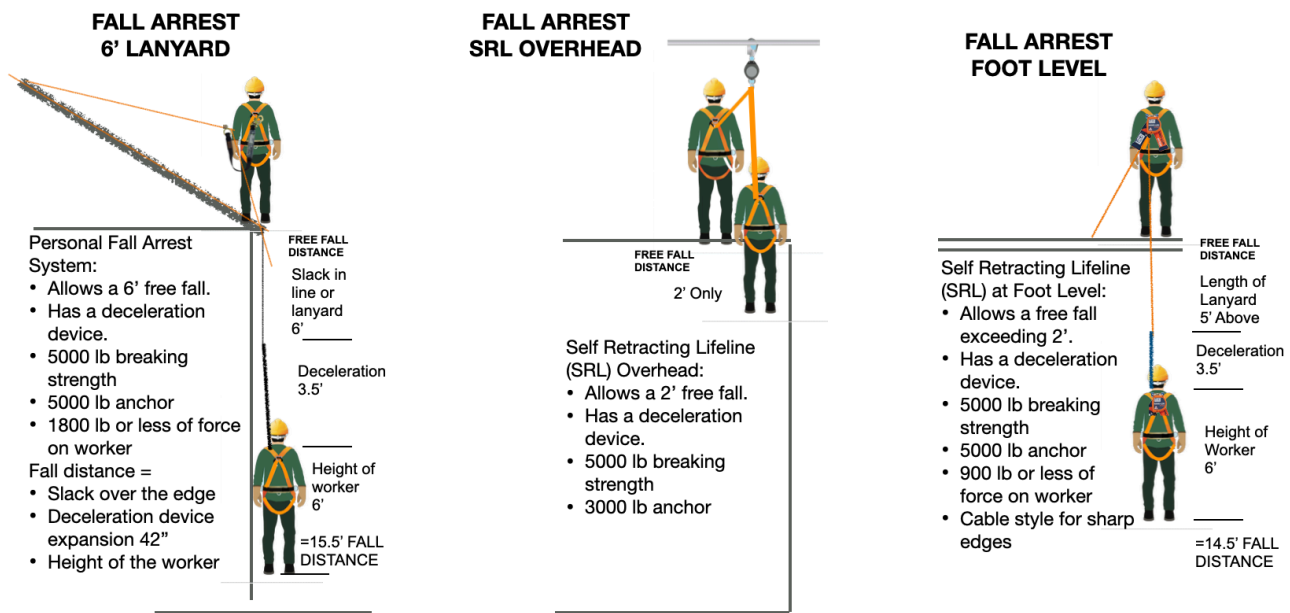
If workers are not lined up with anchors additional slack in the system can be created. This increases fall distance and acceleration. This can translate in

SELF RETRACTING LANYARDS

- If the SRL only allows a free fall of 2' or less, it requires at least 3000 pound anchor.
- If the SRL is used so a free fall exceeding 2' could occur, it must reduce forces to 900 pound or less and it requires a 5000 pound anchor.
- Foot level tie off can result in a 14.5' or more fall distance.

*Note: Many SRLs are marked **Do Not Use** if anchored below dorsal D-ring. Do not use SRLs with nylon straps for foot level tie off, unless designed by the manufacturer.*

FALL DISTANCES

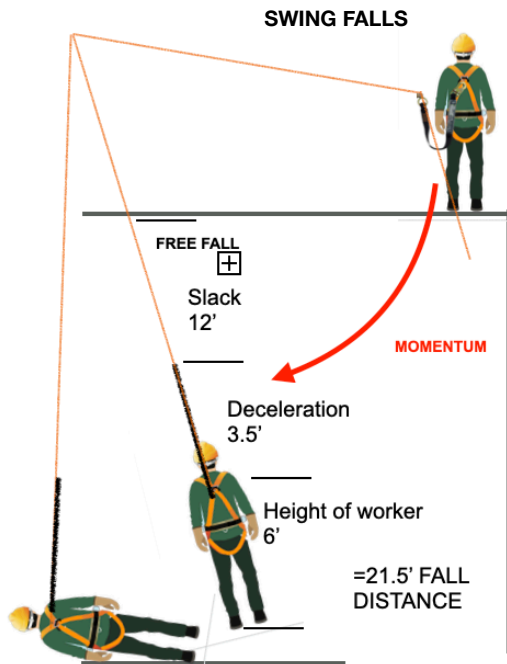


Note: These fall distances are shown without a safety factor.

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severe injuries or death when hitting a wall or the ground. Reduce swing fall by:

- Adding additional anchors
- Account for additional slack in system when calculating fall heights
- Connect to closest vertical anchor.
- Use alternate equipment such as ladders or elevated work platforms.



TOWER CLIMBING FALL ARREST SYSTEMS

Telecommunications work require 100% fall protection at 4'. Examples of 100% tie off on towers is:

- Vertical lifelines (droplines) strength of 5000 lbs in arrest
 - Mainline (Deus or Petzl I'd)
 - Belay (ASAP Absorber)
- Y Lanyards
- Aerial Lifts
- Ladder Climb Safety Systems



INSPECTIONS OF PPE AND FALL PROTECTION EQUIPMENT

- The wearer (employer designated Competent Person) must inspect all PPE before each use.
- Remove any PPE that doesn't pass by marking DO NOT USE or destroying.
- Annual documented inspections are best practice.
- All PPE and equipment must be assembled and maintained per manufacturer's recommendations.

For harnesses, these general recommendations apply:

- Store in clean dry containers, not in sunlight.
- Wash with warm water and soap, bleach free, air dry.
- Additional maintenance and servicing must be by the manufacturer.

Wearer looks for the following:

- Webbing - Cuts, tears, abrasion, fraying, stretching, mold, chemical damage or UV damage
- D-rings - Cracks, breaks corrosion, rough edges, or deformity
- Tongue-buckle - Distortions, added holes, broken grommets

Self Retracting Lanyards (SRLs) Inspection

- SRLs should be stored with the cable fully retracted.
- Always store equipment in a dry location to prevent internal damage to equipment.
- Connecting hardware with impacted load indicators must be removed from service.
- Never attempt to open a SRL, send in for service at the manufacturer.
- SRLs require recertification by manufacturer.

FALL RESCUE

The employer shall provide for prompt rescue of employees in the event of a fall or shall assure that employees are able to rescue themselves.

- Suspension trauma is a concern - most likely due to trauma and shock.
- It is best practice is to never work alone.
- Call 911 immediately, after all significant falls.
- Victim may come down on own power, which is called "Self Rescue"
- Some devices or maneuvers take pressure off femoral arteries, while awaiting rescue (but only work if the victims are conscious). These are not considered Self-Rescue.
- Assisted Rescue includes:
 - Use multiple rescuers to assist.
 - Transfer a worker to a ladder.
 - Transfer worker to a Pulley System or Winch and lower.
 - Use of an Elevated Work Platform.
- Practice your rescue plan!

